

EduForge

MVP Requirements for a School-Focused Web Application

Prepared from the EduForge product idea and role-wise expectations for Principal, Admin Department, Teachers, and Parents.
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Product Focus

Build a lightweight, role-based school operations platform that centralizes student records, attendance, fees, communication, calendar activities, staff visibility, and parent engagement. The MVP should also include visible placeholders for Question Bank and Question Paper Generation so the original EduForge differentiator is present from day one.

Stakeholder Expectations

Role	Expected Outcomes
Principal	Operational overview, staff attendance, teacher leave visibility, fee and salary summaries, payment approvals, student progress, maintenance status, and a concise school health summary.
Admin Department	Student admission records, student profiles, ID cards, fee ledgers, salary/payment tracking, reminders, and payment workflows with Principal approval.
Teachers	Timetable access, student attendance, progress updates, feedback comments, parent comment notifications, homework sharing, question bank, and paper-generation tools.
Parents	Clean child progress view, attendance, homework, upcoming timetable, school activity media, fee visibility, and pickup confirmation through preferred channels.

MVP Requirements

MVP-01. Role-Based Access Control

Purpose: Ensure each user sees only the functions and data relevant to their responsibility.

Primary users: Principal, Admin, Teachers, Parents

- Separate login access for Principal, Admin Department, Teachers, and Parents.
- Parent accounts should only show their own child or children.
- Teachers should only access assigned classes, students, attendance, feedback, and homework.
- Admin should manage records and financial operations, while Principal should have full overview and approval rights.

MVP-02. Principal Dashboard

Purpose: Give the Principal a concise operational control room for the school.

Primary users: Principal

- Show student attendance, staff attendance, fee collection, dues, and key alerts.
- Show teacher leave/time-off overview and student progress highlights.
- Show pending payment approvals and repair/maintenance status.
- Provide a daily or weekly summary of how school operations are running.

MVP-03. Student Admission and Profile Management

Purpose: Create one reliable student database for admissions, profiles, and parent contacts.

Primary users: Admin, Teachers, Principal

- Register new students with class, section, roll number, admission details, and parent/guardian contacts.
- Create and edit student profiles, including basic documents and status.
- Generate a basic student ID card from profile data.
- Support student status such as active, inactive, transferred, or discharged.

MVP-04. Attendance Management

Purpose: Track daily attendance for students, permanent staff, and contractual staff.

Primary users: Teachers, Admin, Principal, Parents

- Allow teachers to mark daily student attendance by class or section.
- Allow Admin or authorized staff to record permanent and temporary staff attendance.
- Show monthly attendance summaries and dashboards for the Principal.
- Send or queue absence notifications for parents.

MVP-05. Fee and Finance Management

Purpose: Track school inflows, outflows, dues, salaries, and approval-controlled payments.

Primary users: Admin, Principal, Parents

- Maintain student fee records, dues, payment history, and fee reminder messages.
- Track salary payments, general income, general expenses, and basic balance sheet summaries.
- Allow Admin to create payment requests and Principal to approve or reject them before discharge.
- Track payment status as pending, approved, rejected, or paid.

MVP-06. Teacher Dashboard

Purpose: Give teachers a practical workspace for classroom operations.

Primary users: Teachers

- Show assigned classes, class timetable, and upcoming classroom activities.
- Allow teachers to mark attendance and record daily, weekly, or monthly student progress.
- Allow teachers to add feedback comments for each student.
- Show parent comments or concerns as notifications.
- Allow teachers to send homework or holiday homework to parents.

MVP-07. Parent Dashboard

Purpose: Give parents a simple, clean view of their child's school life and progress.

Primary users: Parents

- Show attendance, subject-wise academic progress, teacher feedback, and homework.
- Show upcoming timetable, holidays, events, fee dues, and payment status.
- Present progress in a clean infographic-friendly format.
- Show school activity photos or videos to build confidence in extra-curricular focus.

MVP-08. Notifications and Communication

Purpose: Create one communication layer for operational alerts and school-parent updates.

Primary users: Principal, Admin, Teachers, Parents

- Support notifications for fee reminders, attendance alerts, homework updates, teacher feedback, parent comments, announcements, and pickup confirmation.

- Store notification history inside the application.
- Include WhatsApp, SMS, and email as selectable channel placeholders in MVP.
- Actual third-party message integration can be added after the core workflows are stable.

MVP-09. Pickup Tracking

Purpose: Confirm daily student pickup after school close and notify parents.

Primary users: Admin, Teachers, Parents

- Allow authorized school staff to mark a child as picked up.
- Record pickup time and optional pickup person details.
- Send or queue a pickup confirmation to the parent through the selected channel.
- Maintain daily pickup history for audit and reassurance.

MVP-10. Calendar and Activity Management

Purpose: Provide one shared calendar for school events, leaves, exams, holidays, and deadlines.

Primary users: Principal, Admin, Teachers, Parents

- Show upcoming activities in calendar view.
- Support exams, holidays, events, teacher leaves, maintenance schedules, and homework deadlines.
- Allow permitted users to add an activity by selecting a date.
- Control visibility by role so parents see relevant events and staff see operational events.

MVP-11. Maintenance and Repair Tracking

Purpose: Help the Principal monitor repair and maintenance work across the school.

Primary users: Admin, Principal

- Create maintenance tasks with name, description, vendor or assignee, estimated cost, and expected completion date.
- Track task status from open to completed.
- Link payment-related maintenance work to Principal approval.
- Show progress in the Principal dashboard.

MVP-12. Question Bank Placeholder

Purpose: Make the EduForge question-paper differentiator visible in the MVP while keeping scope realistic.

Primary users: Teachers, Principal

- Allow teachers to manually add questions.
- Tag each question by class, subject, chapter or topic, and difficulty level.
- Allow teachers to view and filter saved questions.
- Keep the data structure ready for advanced paper generation later.

MVP-13. Question Paper Generator Placeholder

Purpose: Provide the first visible flow for generating question papers from saved questions.

Primary users: Teachers

- Allow teachers to select class, subject, chapter or topic, total marks, and difficulty mix.
- Provide a Generate Paper action that creates a basic structured preview from available questions.
- Show a clear empty state when there are not enough questions in the bank.
- Keep advanced features such as multiple variants, automatic balancing, and PDF export as phase-two improvements.

MVP-14. Reports

Purpose: Provide basic visibility and accountability across school operations.

Primary users: Principal, Admin, Teachers

- Show reports for student attendance, staff attendance, fee dues, fee collection, student progress, teacher leave, and maintenance status.
- Allow reports to be viewed inside the app in MVP.
- Keep PDF and Excel export as follow-up improvements unless needed immediately.

Recommended MVP Scope

The first version should prioritize operational usefulness and a clear product identity. The following items should be treated as the MVP baseline:

- Role-based login and dashboards.
- Principal dashboard with operational summary.
- Student admission and profile management.
- Student and staff attendance management.
- Fee tracking, dues, reminders, and Principal approval for payment discharge.
- Teacher dashboard for timetable, attendance, progress, feedback, and homework.
- Parent dashboard for progress, attendance, homework, timetable, activity media, and fee status.
- Calendar with activity creation by date.
- Notification history and WhatsApp/SMS/email channel placeholders.
- Pickup tracking with parent confirmation.
- Maintenance and repair tracking.
- Question Bank placeholder.
- Question Paper Generator placeholder.

Out of Scope for First MVP

These are useful, but should be delayed until the core workflows are reliable:

- Advanced AI-based visual teaching aid generation.
- Full WhatsApp/SMS gateway integration, unless required for pilot launch.
- Automated question paper balancing and multiple paper variants.
- Full accounting package replacement with audited financial statements.
- Advanced analytics, predictive insights, and mobile native apps.

Success Criteria

- Principal can understand school operations within one dashboard view.
- Admin can register students, manage fees, prepare payments, and trigger reminders without using Excel as the primary record.
- Teachers can run daily classroom workflows from one dashboard.
- Parents can clearly understand their child's progress, attendance, homework, and pickup status.
- Question Bank and Question Paper Generator are visible enough to validate market interest during early demos.